

		surface is approximately $(0.1 / 0.0) \times (60 \text{ ms kg}^{-1}) = 10 \text{ ms kg}^{-1}$.
11	B	This is an extension from $Q = mc\Delta T$, considering the rate of thermal energy removed. $\frac{dm}{dt} c \Delta T = \frac{dE}{dt} = P$ $\frac{dm}{dt} = \frac{P}{c \Delta T}$ $= \frac{6.7 \times 10^9 / 60}{4200 \times (14.0 - 6.0)} = \frac{6.7 \times 10^9}{4200 \times 8.0 \times 60} \text{ kg s}^{-1}$
12	C	Assume that temperature is constant, then the product of pressure and volume, pV , is